# Mastering Ideas, Not Keystrokes: Digital (3D) Literacy through Digital Humanities Praxis-based Pedagogy

**Schreibman, Susan**
s.schreibman@maastrichtuniversity.nl
Maastricht University, Netherlands, The, DARIAH

**Papadopoulos, Costas**
k.papadopoulos@maastrichtuniversity.nl
Maastricht University, Netherlands, The

**Gilikin Schoueri, Kelly**
k.gillikinschoueri@maastrichtuniversity.nl
Maastricht University, Netherlands, The

Critical Digital Literacy takes many forms, expanding as the field of Digital Humanities expands its methods and theoretical approaches. Critical Digital Literacy moves beyond teaching discipline-specific skills and bespoke tools to incorporating additional competencies, including lifelong, soft, or transferable skills (Wilson et al. 2017). Many of these forward-looking skills are less about learning a specific software package or methods, for example, to conduct interviews, digitise images, or manipulate comma-delimited data, but competencies to empower students to adapt to the ever-changing digital environments in which they will spend the rest of their lives. Situated pedagogic approaches is a backbone of digital humanities praxis-based teaching, with much published research attesting to its value (Augst and Engel 2022; Boulay 2021; Russel and Hensley 2017; Clement 2012). For any given course, some of the soft skills inherent in this approach may be overt (e.g., teaching project management), some may be experiential (learning teamwork by doing). Situated, praxis, based teaching in DH necessarily requires trade-offs –there are only so many skills, competencies, and technologies that can be crammed into a teaching period. If the digital divide may no longer hinge on access to technology, but the ways in which we understand how to use them, how to evaluate what others (including non-human actors) create, how to recognise the real from the hyperreal, hallucinations, or manipulated (Tharib 2023; Khan et al. 2022; Tiernan et al. 2023) , and how to function in team-based technologically-infused environments, then situated teaching can no longer be seen as an add on within our respective domains, but an essential pedagogy for future-based professional and life skills.

This presentation draws from our experience in teaching the Master's level course *Creating Digital Collections* as part of the MA Media Studies: Digital Cultures at Maastricht University. The goal of this research-led course is to provide students with the skills, competencies, and theoretical grounding to develop a collection: an assemblage of digitised physical objects in three-dimensions brought together through an overarching theme that covers their historical, cultural, and social significance. The course has been designed as object-agnostic, therefore, the same principles and core course structure can be applied to different types of objects and consequently different thematic foci. In the academic year 2023-24, students developed the project 'Toys with Voice', a collection of 19 3D scholarly editions, each of which focuses on a digitised toy or plaything. Toys, as cultural artefacts, offer a unique lens through which we can critically examine societal norms, challenge stereotypes, explore notions of gender and social behaviour, scrutinise body image, consider aspects of race, ethnicity, and engage with the impacts of technological innovation and geographical contexts on play.

The course is structured so that the digital collection becomes a repository of narratives, utilising toys as symbolic surrogates to articulate interconnected, multimodal, and non-linear stories. In order to create this collection, several critical literacy skills are taught: photogrammetry (the method used to create the 3D models), multimodality (to develop the multimodal object-based narratives), digital narrative (to write text that can be accessed and read from multiple entry points), digital storytelling (to develop engaging narratives for a broad audience), peer review (how to give critical and constructive feedback to others), and reflection (to critically look at their practice and learning trajectory). These are combined with concepts such as object biography, authenticity, affordances, aura, and digital materiality as well as concepts related to the specific collection, e.g., toys as agents of social change and reflections of social norms.

In this situated approach to course design, we build on Paul Gilster's (1997) definition of digital literacy as "mastering ideas, not keystrokes," while also recognising the increasing expectations of graduates to be competent in an ever-shifting digital and technical landscape (Murray and Perez 2014; Gutiérrez-Ángel et al. 2022). The course objectives are structured across three levels: at

the foundation level, students develop practical skills in using different hardware and software to produce 3D scholarly editions while gaining a theoretical and practical understanding of digital processes and dissemination. At an intermediate level, students critically engage with the impact that design and technical choices have on audiences and stakeholders, and reflect on how their narratives influence audiences and materialise their own situatedness. At the most advanced level, the course revolves around an authentic learning environment (McArthur 2022) by enabling students to apply and extend their knowledge by completing an open-ended task (from the original selection of a toy to the focal points of their narratives and the design of the interaction with the 3D objects) in a real-world context (Wiggins 1990). By situating students in an ongoing research project, PURE3D, a Dutch infrastructure for 3D scholarship led by the authors of this paper, we replicate the conditions and intellectual challenges faced by academics in this field (Koh 2017). These include, for example, the challenges of using a system in production, technical and conceptual issues in developing 3D scholarly editions, as well as managing group dynamics and constraints in resources (e.g. time and the affordances of the software). Such authenticity results in the production of new knowledge; disciplined inquiry and the "aesthetic, utilitarian, or personal value" generated for students (Newmann and Archbald 1992: 72-74). Our students have always expressed their excitement of creating a product that is live and for the public, e.g., contrary to an essay that is written, submitted, and archived, and have learned to appreciate the value of collaboration towards a shared goal.

Our approach to digital, and 3D literacy, emphasises that it cannot be viewed in isolation from other forms of literacy. While this course builds on the strengths and research interests of its developers, the principles of its course design can be transferred to other maker-based approaches. By integrating critical, technical, and reflective practices, we demonstrate that using and mastering digital tools is only one way of creating meaningful engagements with digital environments. By creating 3D scholarly editions and embedding situated and praxis-based teaching within an authentic learning environment, students develop transferable skills that extend beyond discipline specific knowledge. Digital literacy is a multifaceted competency that involves critical making, creative production, and collaborative problem solving, thus preparing students for a competitive digital and creative economy.

# Bibliography


Augst, Thomas / Engel, Deena (2022): "Project-based learning for graduate students in digital humanities", in: Qeios. DOI: 10.32388/4W5TXN

Boulay, Nicole / Morford, Amanda C. / Jacob, Aaron / Patel, Ketan / O'Donnell, Katherine (2021): "Transforming DH pedagogy", in: Digital Studies/le Champ Numérique 11, 1. DOI: 10.16995/dscn.379

Clement, Tanya (2012): "Multiliteracies in the undergraduate digital humanities curriculum: Skills, principles, and habits of mind", in: Hirsch, Brett D. (ed.): Digital humanities pedagogy: Practices, principles and politics. Cambridge: Open Book Publishers, 365–388. https://books.openedition.org/obp/1656?lang=en

Gilster, Paul (1997): Digital literacy. New York: John Wiley.

Gutiérrez-Ángel, Noemí / Sánchez-García, José-Nicolás / Mercader-Rubio, Isabel / García-Martín, Jesús / Brito-Costa, Sónia (2022): "Digital literacy in the university setting: A literature review of empirical studies between 2010 and 2021", in: Frontiers in Psychology 13: 896800. DOI: 10.3389/fpsyg.2022.896800

Khan, Noman / Sarwar, Ayesha / Chen, Thomas B. / Khan, Saira (2022): "Connecting digital literacy in higher education to the 21st-century workforce", in: Knowledge Management & E-Learning 14, 1: 46–61. DOI: 10.34105/j.kmel.2022.14.004

Koh, Kim (2017): "Authentic assessment", in: Oxford research encyclopedia of education. Oxford: Oxford University Press. DOI: 10.1093/acrefore/9780190264093.013.22

McArthur, Jan (2022): "Rethinking authentic assessment: Work, well-being, and society", in: Higher Education. DOI: 10.1007/s10734-022-00822-y

Murray, Matthew C. / Pérez, Juan (2014): "Unraveling the digital literacy paradox: How higher education fails at the fourth literacy", in: Issues in Informing Science and Information Technology 11: 85-100. Retrieved from http://iisit.org/Vol11/IISITv11p085-100Murray0507.pdf

Newmann, Fred M. / Archbald, Douglas A. (1992): "The nature of authentic academic achievement", in: Berlak, Harold et al. (eds.): Toward a new science of educational testing and assessment. Albany: SUNY Press, 71–83.

Russell, Joel / Hensley, Merinda (2017): "Beyond buttonology: Digital humanities, digital pedagogy, and the ACRL framework", in: College & Research Libraries News 78, 11: 588. DOI: 10.5860/crln.78.11.588

Tharib, Salman (2023): "Objective reality in the digital age", in: Metaverse 4, 2. DOI: 10.54517/m.v4i2.2275

Tiernan, Peter / Costello, Eamon / Donlon, Enda / Parysz, Małgorzata / Scriney, Michael (2023): “Information and media literacy in the age of AI: Options for the future”, in: Education Sciences 13, 9: 906. DOI: 10.3390/educsci13090906

Wiggins, Grant (1990): “The case for authentic assessment”, in: Practical Assessment, Research & Evaluation 2, 2. DOI: 10.7275/ffb1-mm19